Department of Electrical and Information Engineering
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Project Report
Embedded Systems
EC6304

Smart Wireless Mesh Sensor Network using Pic Microcontrollers

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17th May 2026

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Abstract

Wireless sensor networks have become an important technology in modern industrial automation, environmental monitoring, disaster management, and agricultural systems. Traditional centralized wireless systems often suffer from communication failures due to signal limitations, node failures, and environmental obstacles. This project proposes the development of a Smart Wireless Mesh Sensor Network using microcontrollers and RF communication modules. The proposed system enables multiple sensor nodes to communicate with each other through a mesh networking approach, ensuring reliable and extended communication.

The system uses ATmega328P microcontrollers together with nRF24L01 RF modules for wireless communication. Environmental sensors such as DHT11, MQ-2 gas sensors, flame sensors, and vibration sensors are integrated into the network to monitor real-time environmental conditions. Sensor data is transmitted across multiple nodes to a central base station for monitoring and alert generation.

The proposed system aims to improve communication reliability, support multi-hop data transmission, and provide real-time monitoring capabilities for industrial and environmental applications.

Contents

Table of Figures

1. Introduction

Wireless communication technologies have significantly improved the ability to collect and monitor data from remote locations. In many applications such as industrial monitoring, agriculture, smart cities, and disaster management, sensor nodes are distributed over a large geographical area. These sensor nodes must communicate efficiently to transmit environmental data to a monitoring center.

Traditional star topology networks depend on a single central node. When the central node fails or goes out of range, communication is interrupted. Mesh networking provides a better solution because each node can communicate with neighboring nodes and forward data through multiple paths. This increases communication reliability and network coverage.

This project focuses on developing a smart wireless mesh sensor network using microcontrollers and RF communication modules. The system is designed to monitor temperature, humidity, smoke, flame, and vibration conditions in real time.

2. Problem Statement

In many real-world environments, environmental monitoring systems face several communication challenges. Sensor nodes placed over large distances often experience signal attenuation, communication interference, and node failures. Traditional point-to-point communication systems cannot guarantee reliable communication in such situations.

Industrial environments may contain physical obstacles that block wireless signals. Agricultural fields may require sensor deployment over very large areas. Disaster-prone regions require continuous and stable communication even when some nodes fail

Therefore, a wireless mesh sensor network is required to improve communication reliability, extend transmission range, and support stable multi-node data transfer.

3. Objectives

3.1 Main Objective

To design and implement a smart wireless mesh sensor network capable of collecting environmental data from distributed sensor nodes and transmitting it reliably to a central monitoring station.

3.2 Specific Objectives

- Enable communication between multiple sensor nodes.
- Extend communication range through multi-hop transmission.
- Collect real-time environmental data including temperature, humidity, gas leakage, flame, and vibration.
- Transmit data wirelessly to a base station for monitoring.
- Generate alerts when abnormal conditions are detected.
- Ensure reliable communication and fault tolerance within the network.

4. Literature Review

Wireless Sensor Networks (WSNs) have been widely studied for applications involving remote monitoring and automation. Researchers have proposed different networking topologies such as star, tree, and mesh networks. Among these topologies, mesh networks provide better fault tolerance and scalability.

Previous studies have shown that mesh networking can improve communication reliability because data packets can travel through multiple nodes before reaching the destination. Several microcontroller platforms such as Arduino, PIC microcontrollers, and ESP32 have been used for WSN applications.

RF modules like nRF24L01 are commonly used due to their low power consumption, low cost, and efficient communication capabilities. Environmental monitoring systems using DHT11 and MQ-2 sensors have also been successfully implemented in smart agriculture and industrial safety applications.

5. Proposed System Architecture

The proposed system consists of multiple wireless sensor nodes connected through a mesh communication network. Each node contains sensors, a microcontroller, and an RF communication module.

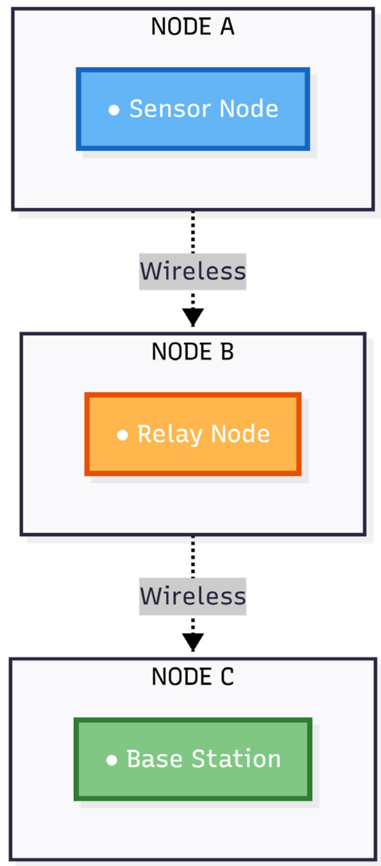

Figure 1:System Architecture

5.1 Input Layer

The input layer includes DHT11 temperature and humidity sensors, MQ-2 gas sensors, flame sensors, and vibration sensors. These sensors continuously collect environmental data from the surroundings.

5.2 Processing Layer

The processing layer is managed by the ATmega328P microcontroller. The microcontroller reads sensor data, converts it into digital values, processes incoming packets, and forwards data to neighboring nodes.

5.3 Output Layer

The output layer consists of a base station connected to an LCD display or PC. The received data is displayed for monitoring purposes, and warning alerts are generated using buzzers or LEDs.

6. Hardware Components and Circuit Design

Component	Function
ATmega328P Microcontroller	Acts as the central processing unit for each sensor node.
nRF24L01 RF Module	Provides wireless communication between nodes using 2.4 GHz frequency.
DHT11 Sensor	Measures temperature and humidity.
MQ-2 Sensor	Detects smoke and gas leakage.
Flame Sensor	Detects the presence of fire.
Vibration Sensor	Detects shocks and vibrations.
16x2 LCD Display	Displays sensor readings and alerts.
AMS1117 Regulator	Provides stable 3.3V power to RF modules.
Buzzer and LEDs	Generate warning signals for abnormal conditions.

6.1 Node A and B

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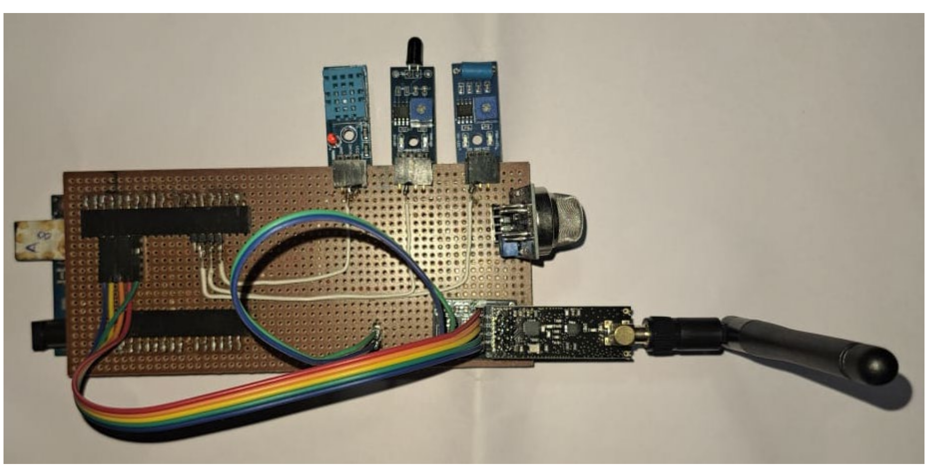

Figure 2:Node A and B

6.2 Node C (Base Unit)

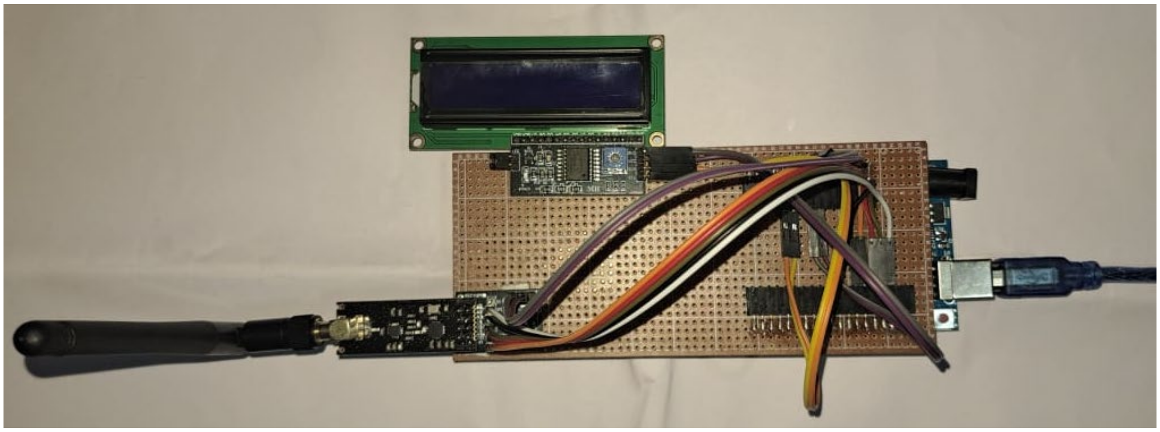

Figure 3:Node C (Base Unit)

6.3 Overall Design

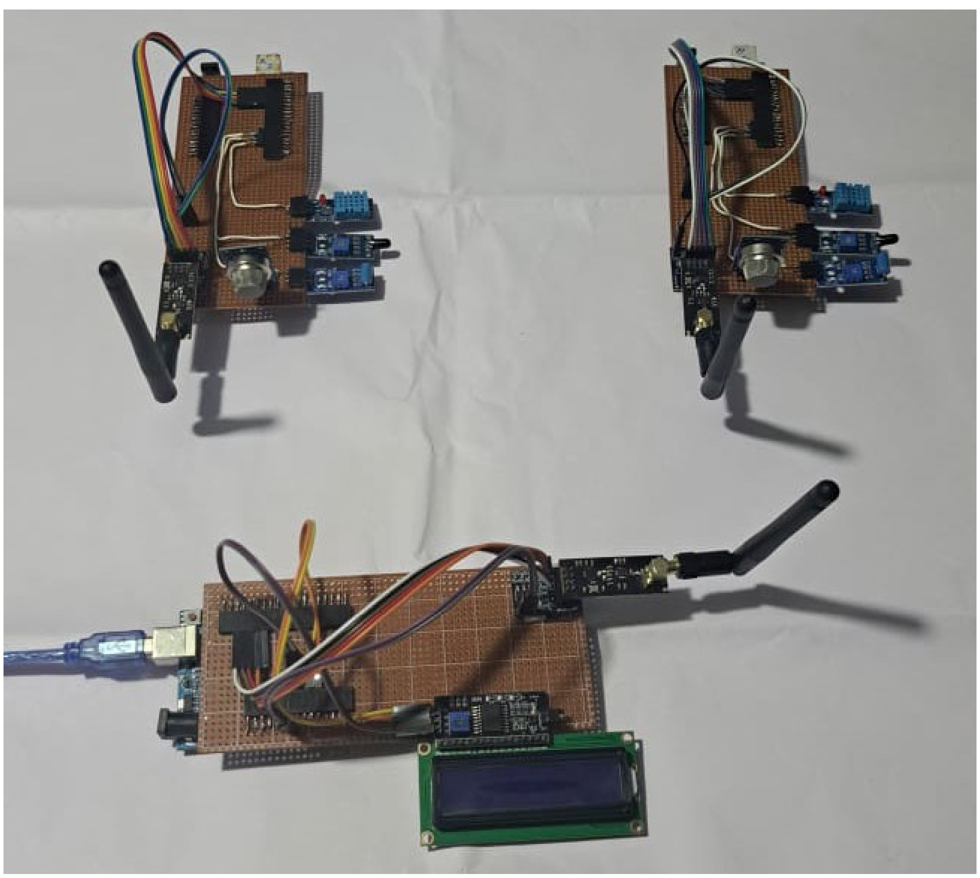

Figure 4:Overall Design

7. Software Design and Flowchart

The software system is programmed using Embedded C within the Arduino IDE environment. The software continuously reads sensor data, processes packets, and forwards data through the mesh network.

1. Initialize sensors and RF communication modules.
2. Read sensor data from connected sensors.
3. Check for incoming packets from neighboring nodes.
4. Process received packets.
5. Forward packets to the next node.
6. Transmit local node data.
7. Display data and generate alerts at the base station.
8. Repeat continuously.

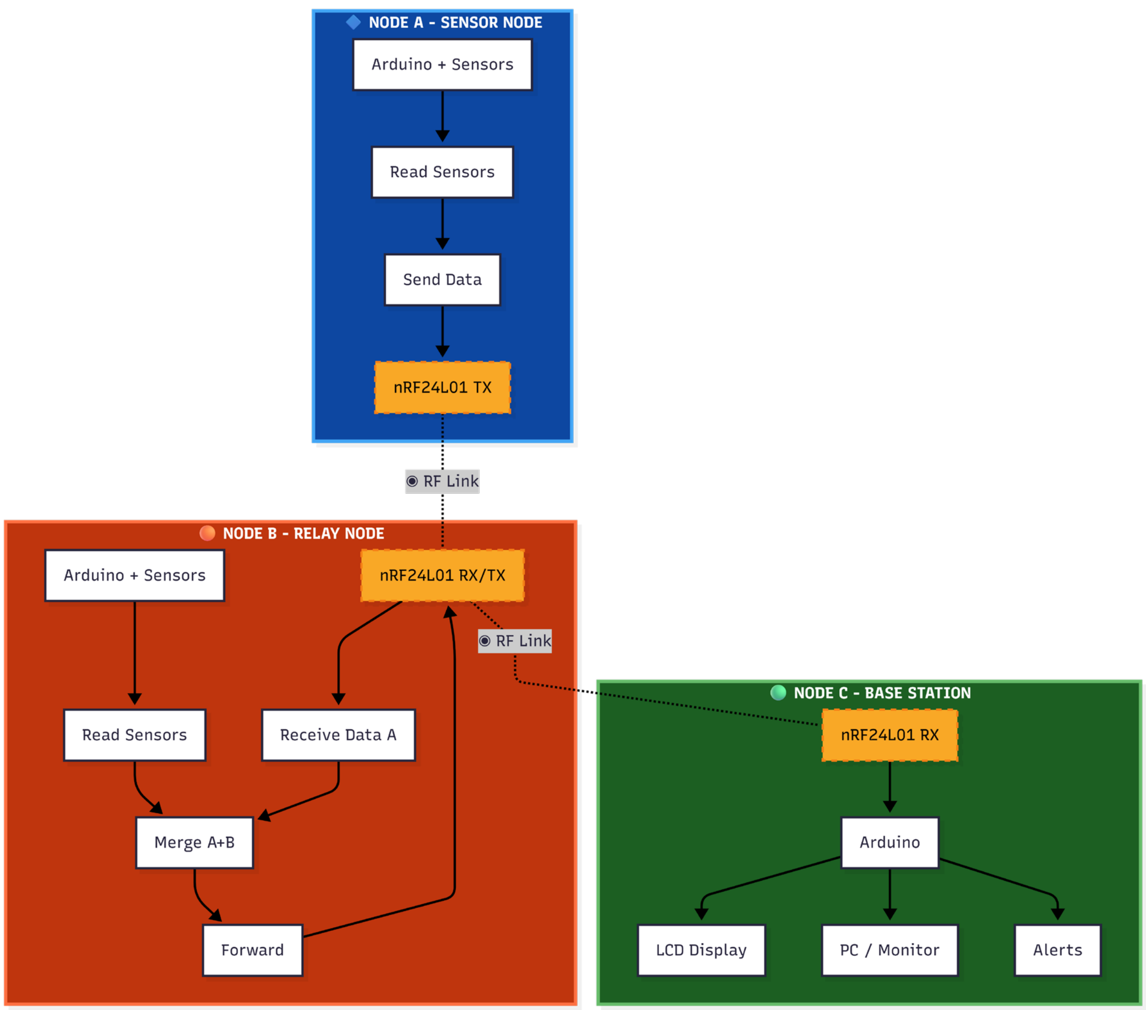

Figure 5:Nodes Flowchart

7.1 Nodes Embedded C Code files

GitHub Repository Link : https://github.com/nethminalakshan/Smart-Wireless-Mesh-Sensor-Network-using-Pic-Microcontrollers.git

8. Methodology

The project implementation begins with designing individual sensor nodes using ATmega328P microcontrollers and environmental sensors. The nodes are connected using nRF24L01 RF communication modules.

Each node is programmed to collect environmental data and transmit it wirelessly through neighboring nodes. A base station is developed to receive and display the transmitted data. Communication tests are conducted to evaluate network reliability and transmission range.

The final system is tested under different environmental conditions to verify sensor accuracy and communication stability.

9. Expected Results and Applications

The proposed system is expected to provide reliable environmental monitoring through wireless mesh communication. The system can support real-time monitoring of temperature, humidity, smoke, fire, and vibration.

Applications of the system include industrial safety monitoring, agricultural environment monitoring, smart buildings, disaster management systems, and security monitoring applications.

10. Advantages and Limitations

10.1 Advantages

- Improved communication reliability.
- Extended communication range.
- Fault-tolerant network structure.
- Low-cost implementation.
- Scalable network design.

10.2 Limitations

- RF communication may experience interference.
- Power consumption increases with multiple nodes.
- Limited transmission speed compared to advanced wireless technologies.
- Environmental obstacles may affect communication quality.

11. Conclusion

The Smart Wireless Mesh Sensor Network is a reliable solution for modern environmental monitoring systems. By combining microcontrollers, wireless RF communication, and multiple environmental sensors, the proposed system provides stable communication, extended coverage, and fault tolerance.

The project demonstrates how mesh networking can improve data transmission reliability in industrial and environmental applications. The proposed system can be expanded further using IoT technologies, cloud connectivity, and advanced sensor integration.

12. References

1. *H. Karl and A. Willig, Protocols and Architectures for Wireless Sensor Networks. Hoboken, NJ, USA: Wiley, 2005.*
2. *Arduino, "ATmega328P and Arduino Uno documentation." [Online]. Available: Arduino official documentation. [Accessed: 17-May-2026].*
3. *Nordic Semiconductor, "nRF24L01 Product Specification." [Online]. Available: Nordic Semiconductor official documentation. [Accessed: 17-May-2026].*
4. *"DHT11 Temperature and Humidity Sensor Datasheet." [Online]. Available: manufacturer datasheet. [Accessed: 17-May-2026].*
5. *"MQ-2 Gas Sensor Datasheet." [Online]. Available: manufacturer datasheet. [Accessed: 17-May-2026].*